# Dan Marshall

Senior full-stack engineer with 11+ years of delivering enterprise .NET and JVM systems. Deep expertise in microservices, event-driven architecture, production AI integration, AI-assisted delivery systems, and developer tooling. Strong advocate for TDD, pair programming, and collaborative engineering practices that improve code quality and team velocity.

**B.Sc. Mathematics and Computer Science** — University of South Africa (UNISA)

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One-page resume available for a concise PDF-friendly version.

Selected public projects are available as a short portfolio reference for technical reviewers.

## Skills

### Back-end Development

| Area | Details |
|---|---|
| **C# / .NET** | Expert knowledge, .NET Framework 3.5 through .NET 11 |
| **JVM Stack** | Spring Boot, Kotlin, Java |
| **Architecture** | Microservices, event-driven architecture, CQRS, DDD, Clean Architecture |
| **Azure** | App Services, Functions, Service Bus, Azure SQL, Blob Storage, App Insights, Entra, Power Platform |
| **AWS** | Bedrock, ECS, SQS, S3 |
| **Testing** | NUnit, xUnit, MSTest 2, JUnit, Selenium, Storybook, TDD, mutation and integration testing |
| **API Development** | ASP.NET WebAPI, FastEndpoints, GraphQL, Spring Boot |
| **Messaging** | MassTransit with Azure Service Bus and event streaming |
| **Data Access** | Entity Framework Core, Dapper, raw SQL |
| **Databases** | SQL Server, PostgreSQL, MySQL, Cosmos DB |
| **Security** | ASP.NET Identity, OAuth2, JWT, Microsoft.Identity, Azure AD / Entra ID |
| **Libraries** | MediatR, Serilog, Polly, AutoMapper, Refit, Quartz.NET, OpenAPI |
| **DevOps** | CI/CD with Azure DevOps, TeamCity, Octopus Deploy, GitHub Pages, PowerShell, Docker |
| **Observability** | OpenTelemetry, Seq, OpenSearch, Application Insights SDK, Serilog Sinks |
| **Static Sites** | Astro, content collections, RSS, metadata validation, image optimisation |
| **Developer Tooling** | Roslyn semantic analysis/refactoring, CLI design, backlog automation, scripted workflows |

## AI/ML Integration

| Area | Details |
|---|---|
| **Production AI Systems** | OpenAI Platform, Claude API, LangChain4j, AWS Bedrock, MCP, agent/evaluation harnesses, prompt engineering |
| **Applications** | Automation, multimodal analysis, guardrails, agentic coding workflows, technical content pipelines |
| **Platforms** | AWS Bedrock, OpenAI, local LLMs, Hugging Face Transformers, llama.cpp-compatible runtimes |
| **Developer Tooling** | Claude Code skills, MCP integration, Codex workflows, PRD backlog automation, prompt and workflow optimisation |
| **Media Generation** | ComfyUI, Flux-based image generation, local AI asset pipelines |

## Front-end Development

| Area | Details |
|---|---|
| **Frameworks** | TypeScript/JavaScript with React, Angular, Astro, Vite, Vitest, and ASP.NET MVC |
| **UI Design** | CSS, Tailwind CSS, Material UI, Storybook |

## General

| Area | Details |
|---|---|
| **Version Control** | Git, GitHub, Bitbucket, Azure Repos |
| **Agile Tools** | Jira, Confluence, Azure DevOps Boards, Teams, Slack |
| **Practices** | TDD, pair/mob programming, code review, mentorship |
| **Operating Systems** | Linux, Windows, macOS, iOS, Android |

# Recent Projects and Achievements

- Rebuilt the CFMEU Wage Claims Manager from a proof of concept into a production system deployed in CFMEU's workspace, remediating security flaws, reducing resource usage, achieving 100% test coverage across the domain and application layers, and adding extensive front-end end-to-end testing.
- Built an AI-assisted delivery system around Claude Code/Codex: reusable skills, project memory, PRD backlog management, TDD execution, mutation testing, Jira/Bitbucket workflow automation, and post-session reflection.
- Built a Roslyn-based C# refactoring CLI for semantic solution-wide rename, move, reference lookup, dependency analysis, duplicate detection, and safe refactoring across legacy ProjectX solutions.
- Drove technical analysis for a pre-launch tour booking platform, including Hero API source-of-truth decisions, PII minimisation, Stripe/Hero payment flow diagnosis, EF migration constraints, and Jira/PR delivery workflow.
- Built an AI-assisted publishing workflow for this Astro blog that drafts articles, generates local illustrations, opens browser previews, publishes to GitHub Pages, and prepares LinkedIn posts from a single command.
- Published technical write-ups on local LLM inference, Claude Code/MCP context optimisation, AI workflow design, and financial rounding correctness.
- Created a Windows setup guide and installer script for running a Bonsai 8B 1-bit LLM locally with CUDA and an OpenAI-compatible API.
- Improved this site with AVIF/WebP image generation, deferred font loading, Shiki syntax highlighting, RSS support, and Playwright coverage for meta descriptions.

# Employment

## Rapid Circle — rapidcircle.com

**Senior Consultant** | July 2026 – November 2026

- Delivered client engineering work across CFMEU, Siemens, Blackmores, and TimeMon in a consulting environment spanning application development, automation, discovery, and integration.
- Rebuilt the CFMEU Wage Claims Manager from a proof of concept into a production system deployed in CFMEU's workspace, remediating security flaws, optimising resource usage, achieving 100% test coverage across the domain and application layers, and adding extensive front-end end-to-end testing.
- Built the Siemens OMVS WMS proof of concept end to end, translating operational requirements into the complete working system and implementation path.
- Extended and hardened TimeMon's Windows client, including session-event monitoring, privacy-hotkey behaviour, and test-backed semantic refactoring across related C# components.
- Performed Blackmores technical discovery, backlog shaping, and engineering analysis.
- Improved internal engineering workflows with AI-assisted tooling, automation, and .NET/Roslyn experiments.

## Reapit — reapit.com

**Senior AI Engineer** | September 2025 – February 2026

- Workshopped, built, and deployed an AI-driven property inspection system using AWS Bedrock, generating structured assessment reports constrained by business rules.
- Forked the LangChain4j framework to add Bedrock guardrail support when unavailable upstream. Modular architecture enabled a seamless swap to the official implementation within one day when released.
- Optimised LLM costs through strategic model selection balancing accuracy, features, performance, and token usage.
- Advised cross-functional teams on practical AI use cases.
- Self-directed adoption of the Kotlin/Spring Boot/AWS ecosystem, delivering production features while supporting the team with JVM implementation patterns.

## RACT (Royal Automobile Club of Tasmania) — ract.com.au

**Senior Software Engineer – Back-end** | June 2024 – July 2025

- Applied Clean Architecture and DDD to enhance testability, scalability, and code clarity.
- Involved in pilot program to develop AI coding rulesets and best practices.
- Migrated and modernised core components from .NET Framework 4.7.2 to .NET 9.
- Built resilient microservices using MassTransit, WebAPI, GraphQL, and SQL Server.
- Collaborated with cross-functional teams to deliver scalable, well-tested features aligned with business requirements.

## National Grower Register — ngr.com.au

**Senior Full-Stack Engineer** | November 2023 – May 2024

- Co-developed a modular online form solution (Flexidocs) using Clean Architecture and DDD, replacing an outdated system.
- Designed modern functional React 18 components with TypeScript and Tailwind CSS.
- Provided guidance on C# programming, OOP, AI tools, and testing best practices.
- Researched and selected sustainable technologies to support long-term goals.
- Improved legacy code with unit test coverage, enabling safe refactoring.
- Contributed to technical documentation and Agile processes.
- Mentored junior staff.

## CloudMonitor — cloudmonitor.ai

**Lead Azure / Back-end Engineer** | June 2023 – November 2023

- Conceptualised, developed, and maintained features running on a serverless platform.
- Promoted TDD, SOLID principles, Clean Architecture, and peer programming.
- Introduced Agile processes, leading morning standups and mentoring junior developers.
- Azure DevOps pipeline maintenance and troubleshooting.
- Introduced Entity Framework, OpenAPI, xUnit, and JetBrains Rider.
- Presented company-wide tech talks and engaged with external stakeholders.

## VALD — vald.com

**Software Engineer – Back-end** | January 2023 – May 2023

- Maintained, refactored, and contributed to various legacy REST APIs and microservices.
- Contributed to the integration testing framework.
- Promoted TDD and peer programming.
- Chaired and contributed to various Agile ceremonies.

## Situ Systems — situsystems.com

**Senior Software Engineer – Full-Stack** | November 2021 – October 2022

- Authored numerous front-end and back-end features.
- Substantially increased test coverage and improved testing methodologies.
- Tuned SQL queries, dramatically improving performance in critical reports.
- Used Pulumi to automate deployment of infrastructure.
- Implemented CI workflows in Azure DevOps.
- Contributed to internal development tools.

## HUB24 Limited — hub24.com.au

**Data and Full-Stack Engineer** | December 2019 – November 2021

- Authored a sophisticated and successful system for cleaning, transforming, and validating fund transfers between superannuation funds.
- Maintained strong working relationships with DBAs, architects, BAs, clients (e.g. Bank of New York), and senior management.
- Provided emergency on-call support across the organisation and managed client infrastructure, triaging and fixing or escalating issues to meet strict SLAs.
- Updated legacy web applications to deploy using Azure Pipelines.
- Championed TDD and automated integration testing.
- Authored, maintained, and supported numerous APIs and ETL projects.

## Chandler Personalised Communications — chandler.com.au

**Data and Full-Stack Engineer** | March 2016 – December 2019

- Worked within an ISO 27001 certified organisation meeting strict security standards.
- Built and maintained web applications servicing government, public utility, and financial sectors.
- Web design using ASP.NET, Bootstrap, CSS, jQuery, and JavaScript.
- Database design in Azure SQL Server.
- Authored SSIS ETL packages to cleanse and process superannuation member data for web, print, and email campaigns.
- Liaised with technical and non-technical stakeholders to develop bespoke solutions.

### WorldSmart POS Solutions — worldsmart.com.au

**Software Engineer, QA and Support** | February 2011 – February 2016

- Developed and maintained enterprise software for retail groups.
- Integrated financial, CRM, and stock modules with MYOB.
- Provided QA via automated scripting and manual testing.
- Authored technical and end-user documentation and provided internal training.
- Maintained internal support utilities and automated scripts.
- Installed and supported Point of Sale hardware and software.

## References

Available on request.

> *“Dan is an absolute pleasure to work with and I would highly recommend him for any full-stack web development position. We worked together for just under a year and during that time Dan was always friendly, professional and went above and beyond when implementing code and designing systems. He has a keen eye for where things can be improved and is always looking to find the optimal solution to any problem he faces. He writes clean, well-structured code that is easy to maintain. He takes advice and constructive criticism well and brings a positive attitude to every meeting. Dan would be an invaluable addition to any software team.”*
>
> — ***Alistair Doulin**, CTO, Situ Systems Pty Ltd*